5 Modeling Either-Or Constraints

Suppose that we have a situation where at least one of the two constraints

$$\boldsymbol{a}_1^T \boldsymbol{x} \leq b_1$$

and

$$\boldsymbol{a}_2^T \boldsymbol{x} \leq b_2$$

is satisfied. We would like to model this either-or condition while preserving the linear structure of the problem. Let M_1 and M_2 be two large enough numbers such that

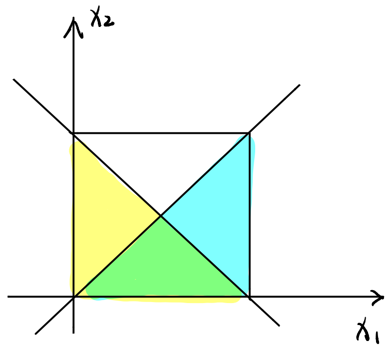
$$\boldsymbol{a}_i^T \boldsymbol{x} \leq b_i + M_i, \quad i = 1, 2$$

for all feasible points in the problem. We can then introduce a binary variable y and model this situation as

$$\begin{aligned} \boldsymbol{a}_1^T \boldsymbol{x} &\leq b_1 + M_1 y \\ \boldsymbol{a}_2^T \boldsymbol{x} &\leq b_2 + M_2(1 - y) \\ y &\in \{0, 1\}. \end{aligned}$$

If $y = 0$, then the first constraint is satisfied and the second constraint is redundant as M_2 is large enough. If $y = 1$, then the second constraint is satisfied and the first constraint is redundant as M_1 is large enough.

e.g.



$$\begin{aligned} 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1 \end{aligned}$$

OR
$$\begin{aligned} x_1 + x_2 \leq 1 \\ -x_1 + x_2 \leq 0 \end{aligned}$$

$\iff$

$$\begin{aligned} 0 \leq x_1 &\leq 1 \\ 0 \leq x_2 &\leq 1 \\ x_1 + x_2 &\leq 1 + 2y \\ -x_1 + x_2 &\leq 2(1-y) \\ y &\in \{0,1\} \end{aligned}$$

6 Modeling p Out of m Constraints

The idea of modeling either-or constraints can be generalized to the problem of satisfying at least p out of m constraints. Suppose that at least p out of the following m constraints must be satisfied.

$$\begin{aligned}
\boldsymbol{a}_1^T\boldsymbol{x} &\leq b_1 \\
\boldsymbol{a}_2^T\boldsymbol{x} &\leq b_2 \\
&\vdots \\
\boldsymbol{a}_m^T\boldsymbol{x} &\leq b_m.
\end{aligned}$$

We can introduce a binary variable y_i for each of the m constraints as

$$y_i = \begin{cases} 1, & \text{if constraint } i \text{ is satisfied} \\ 0, & \text{otherwise.} \end{cases}$$

Let M_i, $i = 1, \ldots, m$ be large enough numbers such that

$$\boldsymbol{a}_i^T\boldsymbol{x} \leq b_i + M_i, \quad i = 1, \ldots, m$$

for all feasible points in the problem. The situation can be rewritten as

$$\begin{aligned}
&\boldsymbol{a}_i^T\boldsymbol{x} \leq b_i + M_i(1 - y_i) \\
&\sum_{i=1}^{m} y_i = p \\
&y_i \in \{0, 1\}, \quad i = 1, \ldots, m.
\end{aligned}$$

7 Representing Integer Variables Using Binary Variables

Suppose that y is a variable which can take integer values from 0 to n. We can set up $n+1$ binary variables x_i, $i = 0, \ldots, n$ as

$$x_i = \begin{cases} 1, & \text{if } y = i \\ 0, & \text{otherwise.} \end{cases}$$

Then, the condition $y \in \{0, 1, \ldots, n\}$ can be equivalently represented as

$$\begin{aligned} &\sum_{i=0}^{n} i\, x_i = y \\ &\sum_{i=0}^{n} x_i = 1 \\ &x_i \in \{0, 1\}, \quad i = 0, \ldots, n. \end{aligned}$$

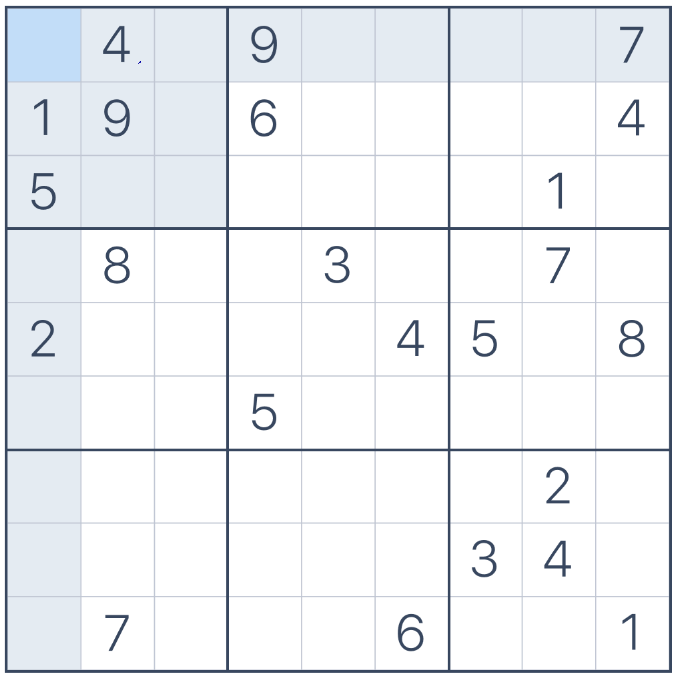
This idea of modeling can be used to represent any variable with a finite number of possible values.

In the above approach, the number of new binary variables is equal to the number of possible values of y. Alternatively, we may have a more compact binary representation of y. Let u be the smallest integer such that $2^u > n$. For example, if $n = 12$, then $u = 4$. We can define u binary variables z_i, $i = 0, \ldots, u-1$ and represent the condition $y \in \{0, 1, \ldots, n\}$ equivalently as

$$\begin{aligned} &\sum_{i=0}^{u-1} 2^i\, z_i = y \\ &y \leq n \\ &z_i \in \{0, 1\}, \quad i = 0, \ldots, u-1. \end{aligned}$$

8 The Sudoku Game

The goal of Sudoku is to fill in a 9×9 grid with digits so that each column, row, and 3×3 section contain the numbers between 1 to 9. At the beginning of the game, the 9×9 grid will have some of the squares filled in. Your job is to use logic to fill in the missing digits and complete the grid.



	4		9					7
1	9		6					4
5							1	
	8			3			7	
2					4	5		8
			5					
							2	
						3	4	
	7				6			1

Figure 1: From: https://sudoku.com

Variables. We define $9 \times 9 \times 9 = 827$ binary variables as follows.

$$x_{ijk} = \begin{cases} 1 & \text{if number } k \text{ is in the } i\text{th row and } j\text{th column} \\ 0 & \text{otherwise.} \end{cases}$$

For example, in the figure above, $x_{124} = 1$.

Model.

$$
\begin{array}{lll}
\min & 0 & \\
\text{s.t.} & \displaystyle\sum_{k=1}^{9} x_{ijk} = 1, & i, j = 1, \ldots, 9 \\
& \displaystyle\sum_{j=1}^{9} x_{ijk} = 1, & i, k = 1, \ldots, 9 \\
& \displaystyle\sum_{i=1}^{9} x_{ijk} = 1, & j, k = 1, \ldots, 9 \\
& \displaystyle\sum_{i=3s-2}^{3s} \sum_{j=3t-2}^{3t} x_{ijk} = 1, & k = 1, \ldots, 9,\ s = 1, 2, 3,\ t = 1, 2, 3 \\
& x_{ijk} = 1, & \text{for all known } (i, j, k) \\
& x_{ijk} \in \{0, 1\}, & i, j, k = 1, \ldots, 9.
\end{array}
$$

The first constraints ensure that exactly one of the nine numbers appear in each cell. The second constraints guarantee that each row contains each number exactly once. Similarly, the third and fourth constraints guarantee that each column and each block contains each number exactly once, respectively. The fifth constraints indicate all known information. The objective function can be set to any constant, as we are simply looking for a feasible solution.